# NOISE POLLUTION AND ITS EFFECTS

Noise pollution refers to excessive or disruptive sound that negatively impacts the environment and human health. Common sources include traffic, industrial activities, construction, and loudspeakers. Prolonged exposure to high noise levels can lead to hearing loss, stress, sleep disturbances, and cardiovascular issues.

It also disrupts wildlife by interfering with communication, navigation, and reproduction. Reducing noise pollution through better urban planning, soundproofing, and stricter regulations is essential for maintaining a healthier and more peaceful environment.

# ESP32-CAM FUNCTIONALITY IN CLASSROOM NOISE DETECTION

1. Noise Detection Trigger:
The sound sensor detects excessive noise levels in the classroom.
If the noise exceeds a set threshold, the ESP32 processes the data.
2. ESP32-CAM Activation:
When noise crosses the permissible limit, the ESP32-CAM captures images or videos of the classroom.
This helps in identifying the source of the noise.
3. Data Transmission via Wi-Fi:
The captured image/video is sent to a remote server, cloud storage, or a mobile device.
Teachers or administrators can review the data remotely.
4. Real-Time Alerts & Notifications:
The system sends alerts to teachers via a mobile app, email, or web dashboard.
A buzzer or LCD display may also indicate high noise levels in real-time.
5. Classroom Monitoring & Behavior Analysis:
Helps teachers track classroom discipline.
Provides a visual record of noisy instances for further review.
Expected Benefits for Classroom Management:
✔ Enhances classroom discipline by discouraging disruptive behavior.
✔ Real-time noise monitoring improves teaching effectiveness.
✔ Remote access allows teachers to monitor multiple classrooms.
✔ Data logging helps analyze noise trends over time.

This system makes classroom noise monitoring more efficient by combining audio detection, image capture, and real-time alerts

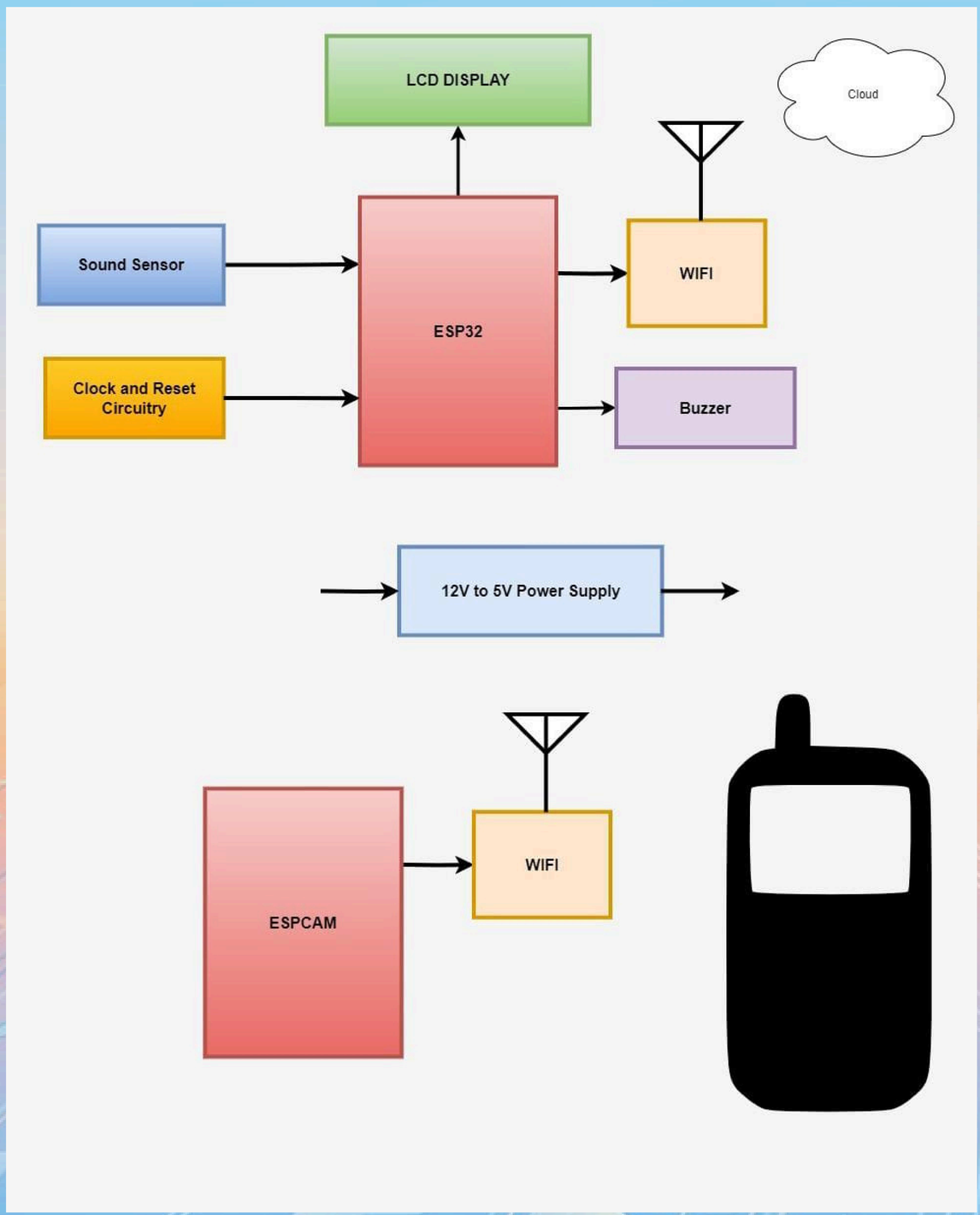
# SCOPE

An **IoT-enabled noise monitoring system** with an alert mechanism in industries helps ensure workplace safety and regulatory compliance by continuously tracking noise levels using smart sensors. It provides real-time data, sends alerts when noise exceeds safe limits, and helps reduce health risks like hearing loss and stress among workers. The system also supports predictive maintenance by identifying high-noise machinery, improving operational efficiency. By integrating with Industrial IoT (IIoT) systems, it enhances workplace safety, regulatory adherence, and environmental sustainability.

# METHODOLOGY

**IoT Sound Level Sensors:** Continuously monitor the environment and measure noise levels in decibels (dB).
**Data Acquisition:** The sensors collect real-time noise data.
**Data Transmission:** The collected data is sent via Wi-Fi to a cloud server or local edge device.
**Data Processing & Predictive Analytics:** Incoming data is analyzed to detect when noise levels exceed the permissible limits and to predict potential issues.
**Automated Alerts:** When thresholds are exceeded, alerts are automatically sent to concerned personnel through SMS, email, or mobile applications.
**Noise Control Actions:** Based on insights, industries can implement control measures like equipment maintenance, soundproofing, or operational adjustments to prevent excessive noise generation, ensuring a safer work environment and regulatory compliance.

# BLOCK DIAGRAM




# APPLICATIONS

**Industrial Safety & Compliance –** Ensures safe noise levels in factories and construction sites while meeting regulations.

**Urban Noise Pollution Control –** Helps monitor and manage noise from traffic, construction, and public activities.

**Smart Cities –** Integrates with city systems for real-time noise mapping and better urban planning.

**Traffic & Transport –** Tracks noise from highways, railways, and airports to aid in soundproofing measures.

**Healthcare & Education –** Maintains quiet environments in hospitals, schools, and libraries for better well-being.

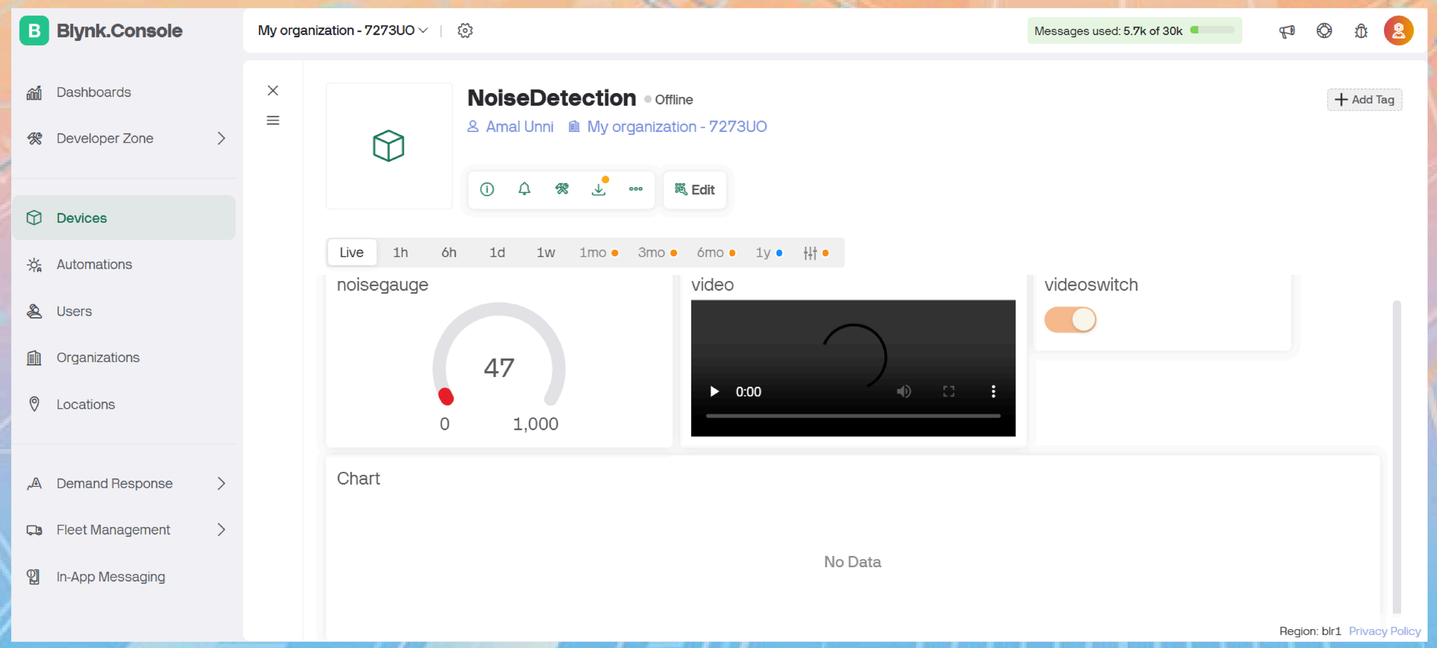
**Residential & Commercial Spaces –** Controls noise in housing societies, offices, and malls to enhance comfort.

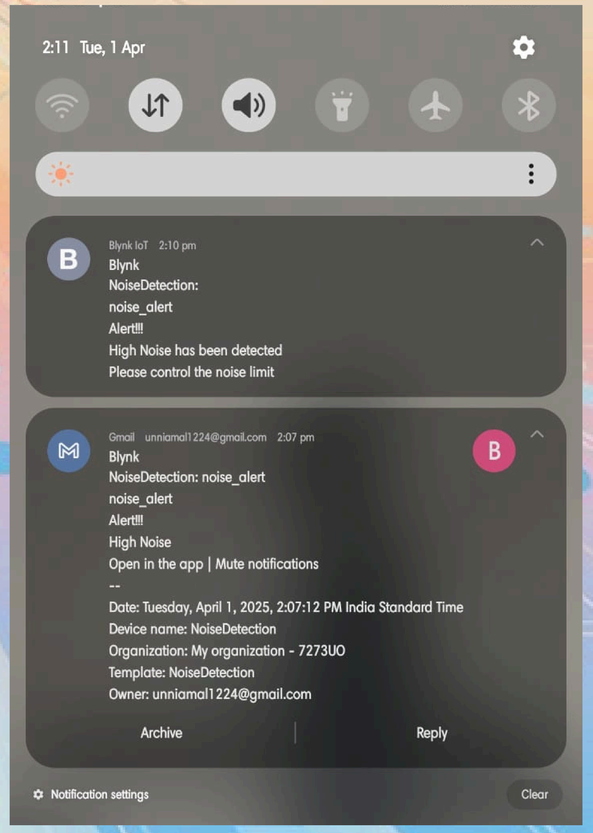
**Wildlife & Environment –** Monitors noise impact on wildlife to support conservation efforts

**Events & Entertainment –** Ensures noise control at concerts and public gatherings to prevent disturbances.

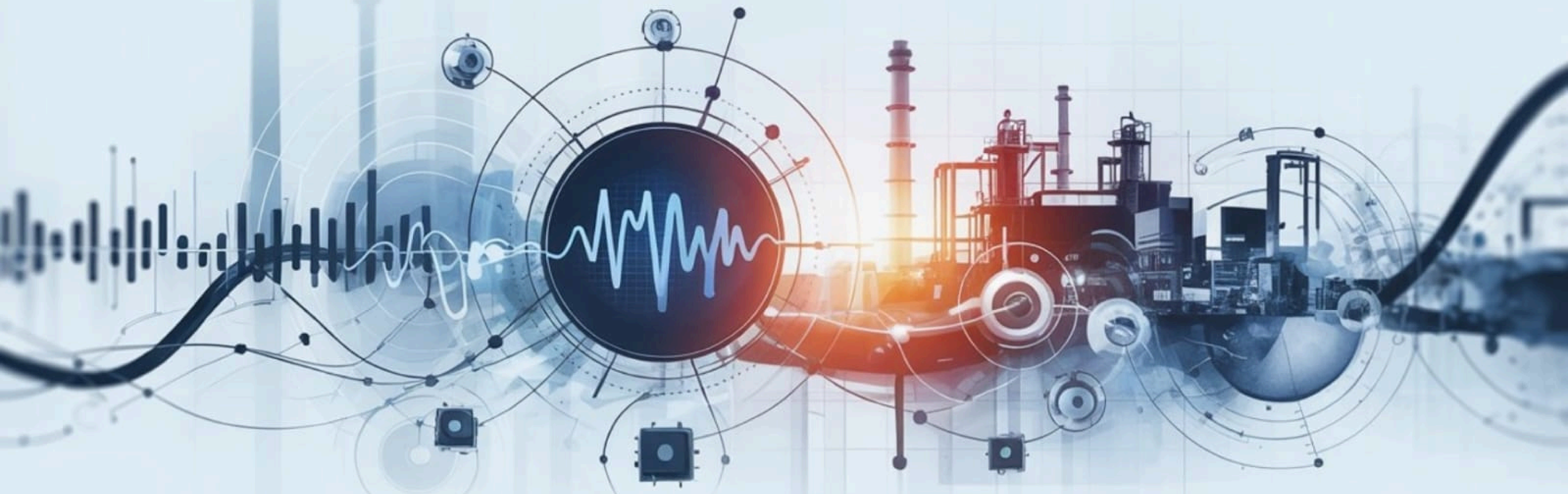
# FINAL RESULTS

The LCD display shows real-time sound levels and intensity. If noise exceeds 75 dB in industrial settings, a buzzer alerts staff and sends a notification via the Blynk app. In classrooms, when noise reaches a threshold, the ESP32-CAM captures and stores an image for review. Additionally, a chart provides a visual representation of noise data over time, aiding trend analysis and prevention.



# IOT ENABLED NOISE MONITORING SYSTEM




**PROJECT BY**

NANDHINI M | AMAL S KUMAR | ANUMOD S. NAIR | AKHIL RICHARD